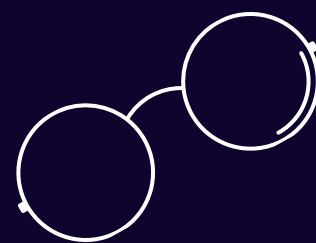
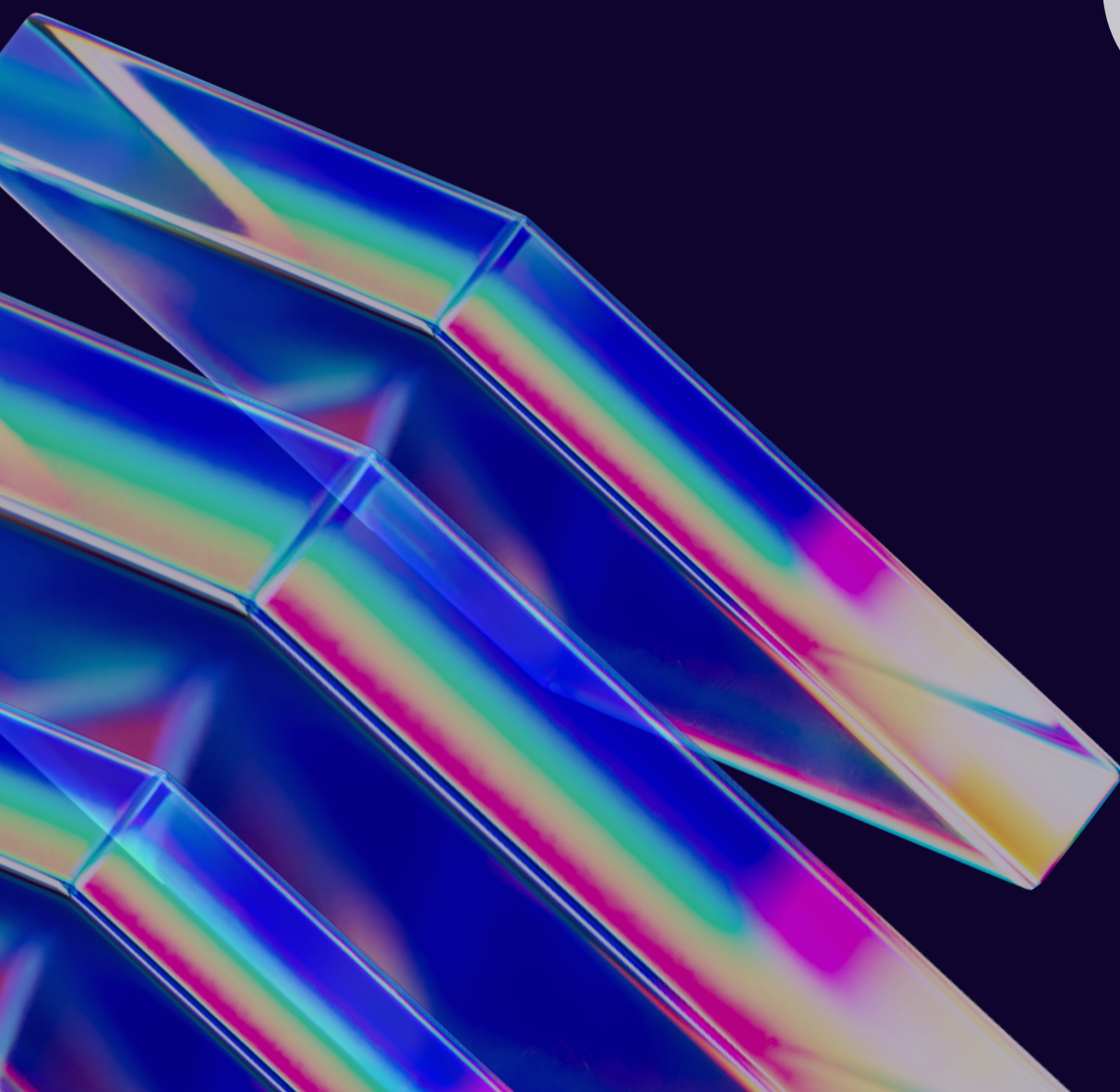


ANTI-EPILEPTIC GLASSES



Problem Statement

Over 50 million people worldwide suffer from epilepsy, with nearly 80% living in low- and middle-income regions. Among them, approximately 3% are affected by Photosensitive Epilepsy, where seizures are triggered by sudden changes in light intensity, flickering lights, or high-contrast visual patterns.

These triggers are commonly encountered in everyday environments such as digital screens, traffic signals, and artificial lighting, making it difficult for affected individuals to avoid exposure. Despite this risk, existing solutions are largely passive—relying on avoidance or static tinted lenses—which do not effectively respond to dynamic lighting conditions.

This highlights a critical need for an affordable, real-time, and adaptive system that can automatically respond to changes in light intensity and provide continuous visual protection in daily life.



SOLUTION APPROACH

Our Solution

We developed a smart wearable system in the form of glasses that provides real-time protection for individuals with Photosensitive Epilepsy.

Key Concept

The system continuously monitors ambient light intensity and automatically adjusts the lens opacity when a sudden increase or fluctuation in light is detected.

Core Features

- Real-time light intensity sensing
- Immediate response to sudden brightness changes
- Automatic lens darkening without user intervention
- Compact and wearable design

Outcome

The system ensures continuous visual protection and reduces exposure to potential seizure-triggering stimuli in everyday environments.

INNOVATION

What Makes Our Project Unique

Real-Time Adaptive Response

Unlike traditional lenses, our system actively detects and responds to sudden changes in light intensity within moments.

Dynamic Lens Adjustment

The lenses darken only when required and return to normal otherwise, ensuring both protection and visibility.

Fully Automatic Operation

No manual control is needed. The system works independently in real time, reducing user effort and reaction delay.

Wearable and Environment-Independent

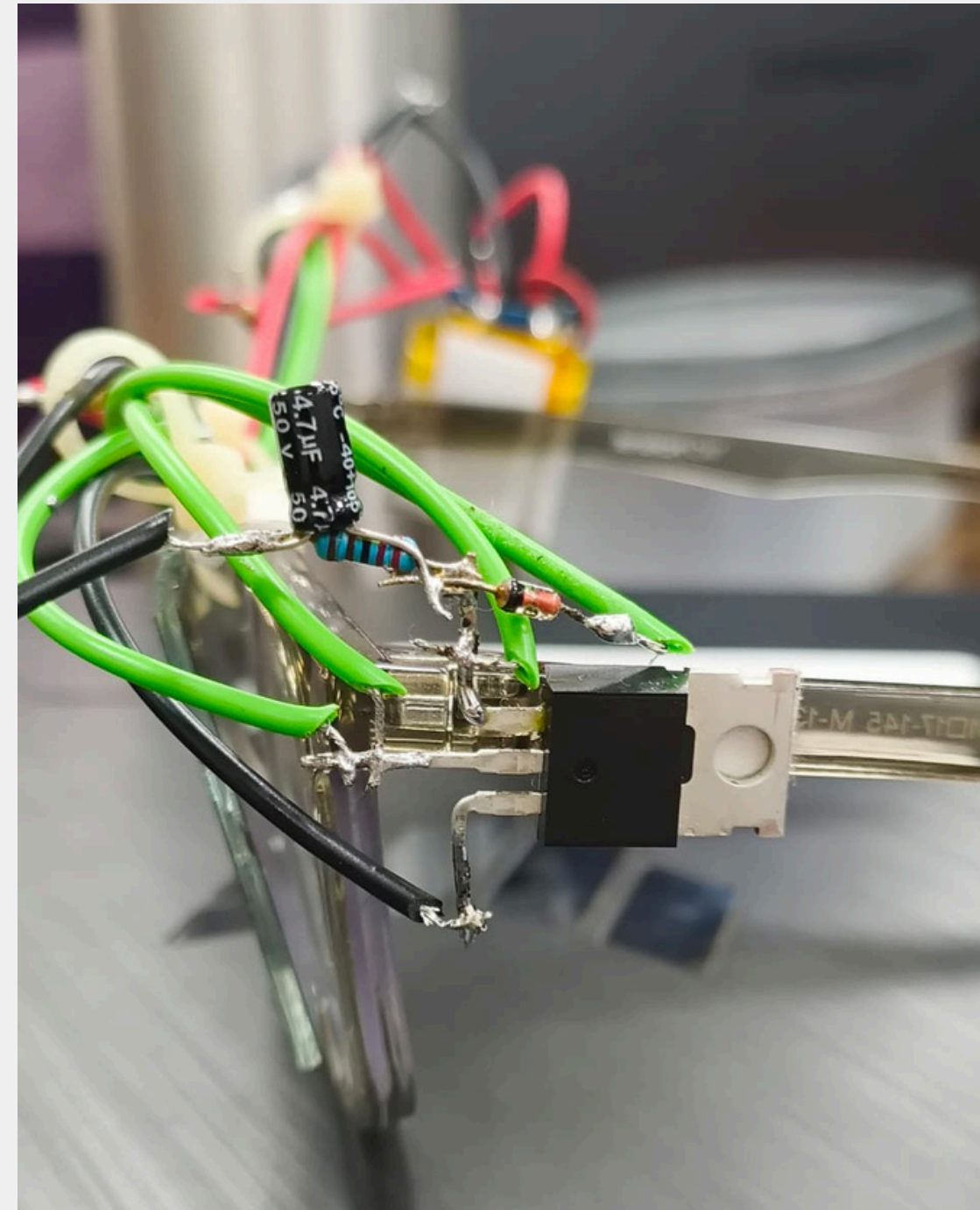
The solution is not limited to screens or specific locations. It works in any environment, including outdoors and public spaces.

Preventive Rather Than Reactive

Instead of responding after exposure, the system acts immediately at the onset of a trigger, reducing seizure risk proactively.

COMPONENTS USED

- MOSFET - IRLZ44N (ACTS AS A SWITCH)
- DIODE - 1N4148 (ENSURES FAST SWITCHING)
- RESISTORS
- CAPACITORS
- LDR(LIGHT DEPENDENT RESISTOR)
- POTENTIOMETER
- BATTERY (3.7V 400mA)
- CHARGING MODULE(TP4056)
- SMALL LIQUID CRYSTAL LIGHT VALVE (LENS)
- CONNECTING WIRES



IMPLEMENTATION

Light Detection

- LDR senses ambient light intensity
- Connected in a voltage divider with a potentiometer
- Potentiometer sets the sensitivity threshold

Signal Processing

- Output voltage varies with light intensity
- Resistors and capacitors used for biasing and noise reduction

Switching Mechanism

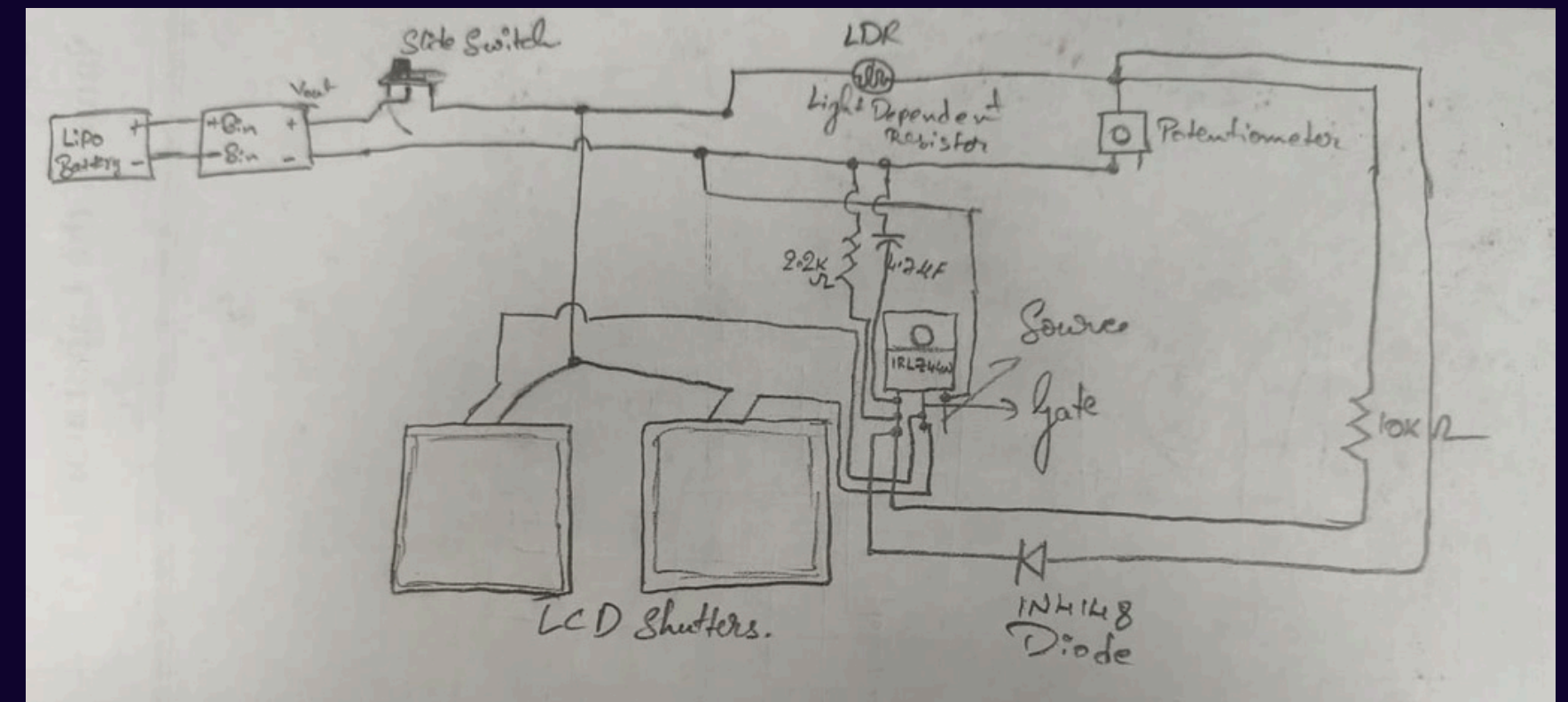
- IRLZ44N MOSFET acts as a switch
- Turns ON when light exceeds threshold
- Turns OFF when light intensity decreases
- 1N4148 diode ensures fast and stable switching

Lens Control

- MOSFET controls power to the liquid crystal lens
- Lens darkens when powered
- Returns to transparent state when power is removed

Power Supply

- Powered by 3.7V battery
- TP4056 module used for safe charging
- Ensures portability for wearable use



ROUGH CIRCUIT DIAGRAM

SCALABILITY

- Low-cost design (~₹900)
- Easily reproducible
- Compact and wearable
- Scalable manufacturing
- Upgrade-friendly system
- Multi-application use

Component-Level Scalability

- Uses low-cost and widely available components
- Can be easily reproduced and assembled in large quantities
- Suitable for cost-effective mass production

Design Flexibility

- Circuit can be miniaturized into a compact PCB
- Can be integrated into different types of eyewear
- Adaptable to various lens technologies

Feature Expansion

- Can be upgraded with microcontrollers for smarter control
- Possible integration of multiple sensors (light, flicker detection)
- Can include adjustable modes for different environments

Manufacturing Potential

- Simple design enables easy transition from prototype to product
- Compatible with standard electronics manufacturing processes

Application Scalability

- Can be extended beyond epilepsy to general eye protection
- Useful for environments with high light fluctuations (industrial, outdoor use)

FEASIBILITY

Technical Feasibility

- Uses simple and proven electronic components
- Easy to design, implement, and troubleshoot
- Reliable real-time response to light changes

Economic Feasibility

- Low-cost components (approx. ₹900 total)
- Affordable for large-scale production
- Cost-effective compared to specialized alternatives

Practical Feasibility

- Compact and wearable design
- Operates automatically without user intervention
- Suitable for everyday use

Operational Feasibility

- Low power consumption
- Rechargeable battery with charging module
- Minimal maintenance required

Implementation Feasibility

- Easy to assemble and integrate
- No complex manufacturing required
- Can be developed with basic electronics knowledge



IMPACT

Health and Safety

- Reduces risk of seizures caused by sudden light changes
- Provides real-time protection for individuals with Photosensitive Epilepsy

Improved Quality of Life

- Enables safer interaction with everyday environments
- Reduces need to constantly avoid screens or bright areas

Accessibility and Affordability

- Low-cost solution (~₹900)
- Accessible to a wider population, especially in low-resource settings

Real-World Usability

- Works in both indoor and outdoor environments
- Fully automatic operation without user intervention

Technological and Social Impact

- Demonstrates practical use of electronics in healthcare
- Encourages development of affordable assistive technologies

QUICK SUMMARY

- Reduced seizure risk
- Increased independence
- Affordable (~₹900)
- Real-time protection
- Everyday usability
- Assistive innovation

PERFORMANCE

Response Time

- Rapid response to sudden changes in light intensity
- Near-instant activation of lens darkening

Sensitivity

- Adjustable sensitivity using potentiometer
- Detects small to significant variations in light

Stability

- Stable operation with minimal noise due to capacitors
- Consistent performance under varying conditions

Power Efficiency

- Low power consumption
- Efficient operation using a 3.7V battery

Reliability

- Consistent switching using MOSFET
- Fast and reliable response ensured by diode

Overall Performance

- Smooth and automatic operation
- Effective real-time protection in different environments



Stable operation



Real-time performance



Fast response



Low power usage



Reliable switching



Adjustable sensitivity

FUTURE SCOPE

Advanced Sensing

- Integration of flicker detection in addition to light intensity
- Improved accuracy in identifying seizure-triggering patterns

Smart Control System

- Use of microcontrollers for adaptive and programmable response
- Automatic adjustment based on environment and user preference

Miniaturization

- Conversion into a compact PCB for better portability
- **Improved** design for seamless integration into regular eyewear

Enhanced Lens Technology

- Use of faster and more efficient liquid crystal or electrochromic lenses
- Better transparency control and response time

User Customization

- Adjustable sensitivity levels for different users
- Possible mobile app interface for control and monitoring

Wider Applications

- Use in industrial safety, driving, and outdoor protection
- Can be extended to general eye protection and vision safety

THANK YOU